

SHAHAREYAR HOSSAIN

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Education

Ramakrishna Mission Vivekananda Educational and Research Institute

Aug 2022 – May 2024

Masters of Science in Data Science - Gold Medalist - CGPA: 9.2

Howrah, West Bengal

University of Calcutta

Aug 2018 – June 2020

Masters of Science in Pure Mathematics - CGPA: 7.02

Kolkata, West Bengal

Experience

Cloudcraftz Solutions Pvt Ltd

June 2024 – Present

Data Scientist

Kolkata, West Bengal

- Developed a predictive model for market prices of commodities (gold, silver, copper) based on market sentiments.
- Collected and processed news articles and social media data (Twitter, Reddit, Google) using **NewsApi**, **Twitter Api** and **Reddit Api**.
- Implemented **OpenAI** relevance and sentiment analysis pipelines to assess the impact of news and social media on commodity prices.
- Developed a trading strategy informed by sentiment trends.
- Improved the accuracy of price predictions by incorporating sentiment-driven insights into commodity market forecasting.

SenSight Technologies Pvt Ltd

Jan 2024 – May 2024

Data Scientist Intern

Bengaluru, Karnataka

- Developed an LLM-powered chatbot using the **RAG** (Retrieval-Augmented Generation) framework to refer to external data sources for generating accurate answers.
- Implemented **Microsoft's Table Transformers** to detect tables present in documents, enhancing document processing capabilities.
- Utilized open-source embeddings models to create vector embeddings of documents, enabling semantic analysis and similarity search.
- Developed and implemented a **Hybrid search** system, combining traditional keyword search with vector-based similarity search, for improved document retrieval accuracy.
- Stored document embeddings on vector databases like **Qdrant** and **Chroma** for efficient retrieval and management of large document datasets.
- Built a user-friendly web application using **Streamlit** for easy interaction with the chatbot.

Projects

Fine-Tuning Language Model for Domain Expertise | *Generative AI, Deep Learning, Pytorch*

August 2023

- Led a project focused on fine-tuning the Google's **FLAN-T5** model for dialogue summarization tasks.
- Curated and preprocessed the DialogSum dataset for training and evaluation purposes.
- Implemented advanced techniques such as **LoRA**, **QLoRA** to fine-tune the FLAN-T5 architecture for less GPU memory requirement.
- Trained only **0.71%** of original model parameters using **PEFT** techniques.
- Used **ROUGE** Score for Model evaluation. Absolute percentage improvement of PEFT Model over Original Model:
Rouge1: 80.6%, Rouge2: 60.5%, RougeL: 66.8%

Technical Skills

Programming Skills:

- Python, R, C, SQL

Data Science:

- Machine Learning, Deep Learning, NLP, Large Language Models, Generative AI, Statistical Analysis, Time Series Forecasting

Technologies/Frameworks:

- GitHub, Django, REST API, Streamlit, HTML, Reflex, HuggingFace, OpenAI, Prophet, Langchain, Tensorflow, Pytorch